



UNIVERSITY OF BURGUNDY

POLYTECH DIJON

Path Planning on Grid Map

PRM and RRT Comparison

Report

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A concise comparison between PRM and RRT using visual and numerical results.

Introduction

In this work, we implemented and compared two classical path planning algorithms: PRM (Probabilistic Roadmap) and RRT (Rapidly-exploring Random Tree). The objective was to study how they behave in the same 2D environment with obstacles, and then compare them in terms of execution time and path quality.

Exercise 1: Environment

We first created a 2D environment containing several obstacles, one start point and one goal point. This environment is used for both algorithms in order to make the comparison fair and clear.



Figure 1: 2D environment with obstacles, start point in green and goal point in red

Exercise 2: PRM

The PRM method starts by generating random collision-free points in the workspace. Then, nearby points are connected to build a roadmap. Once the graph is created, the shortest path between the start and the goal is extracted.

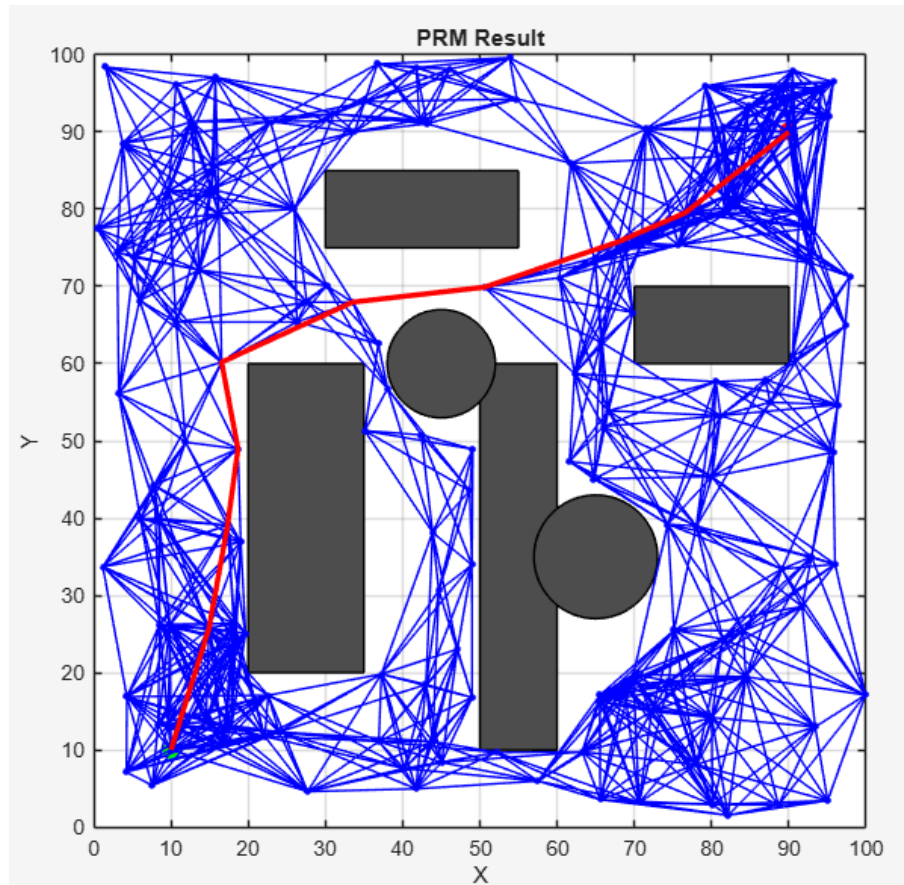


Figure 2: PRM result

The result shows that PRM creates a dense graph in the free space. The final path is relatively short and smooth, which is one of the main advantages of this method.

Exercise 3: RRT

The RRT method builds a tree progressively from the start point. At each step, a random point is sampled and the tree extends toward it if the motion is collision-free.

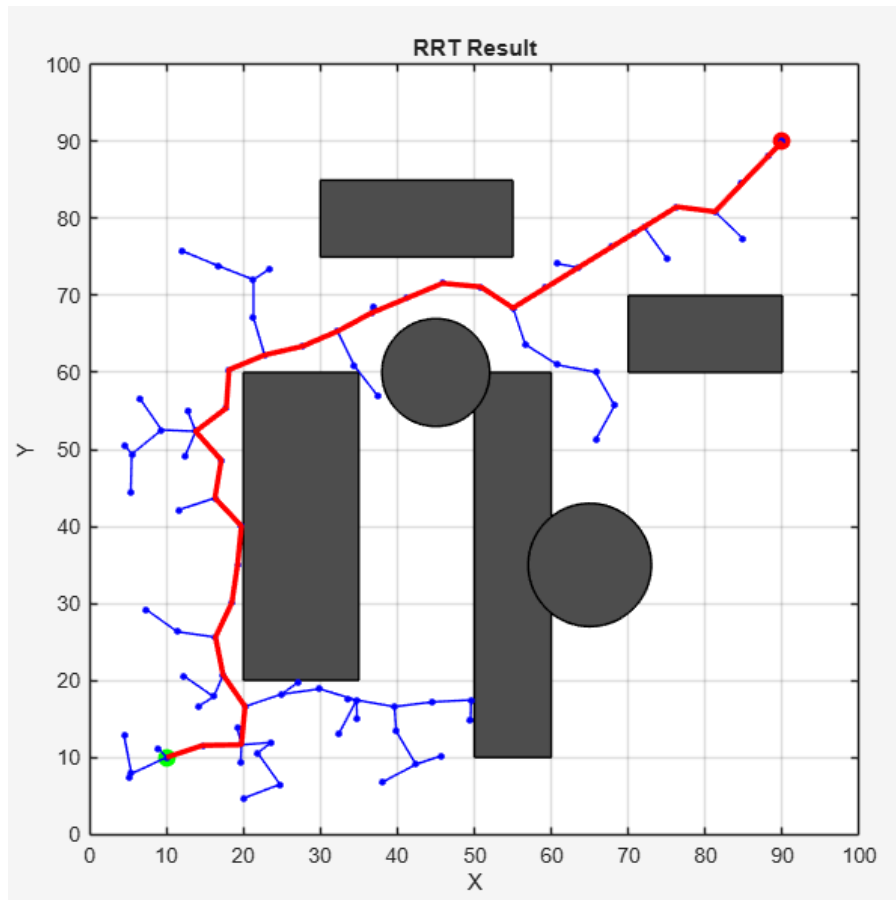


Figure 3: RRT result

We can clearly see the exploration behavior of the tree. A valid path is found, but it is less direct than the one obtained with PRM.

Exercise 4: Comparison Between PRM and RRT

To compare both methods, we varied the number of nodes and observed the effect on execution time and path length.

Execution Time

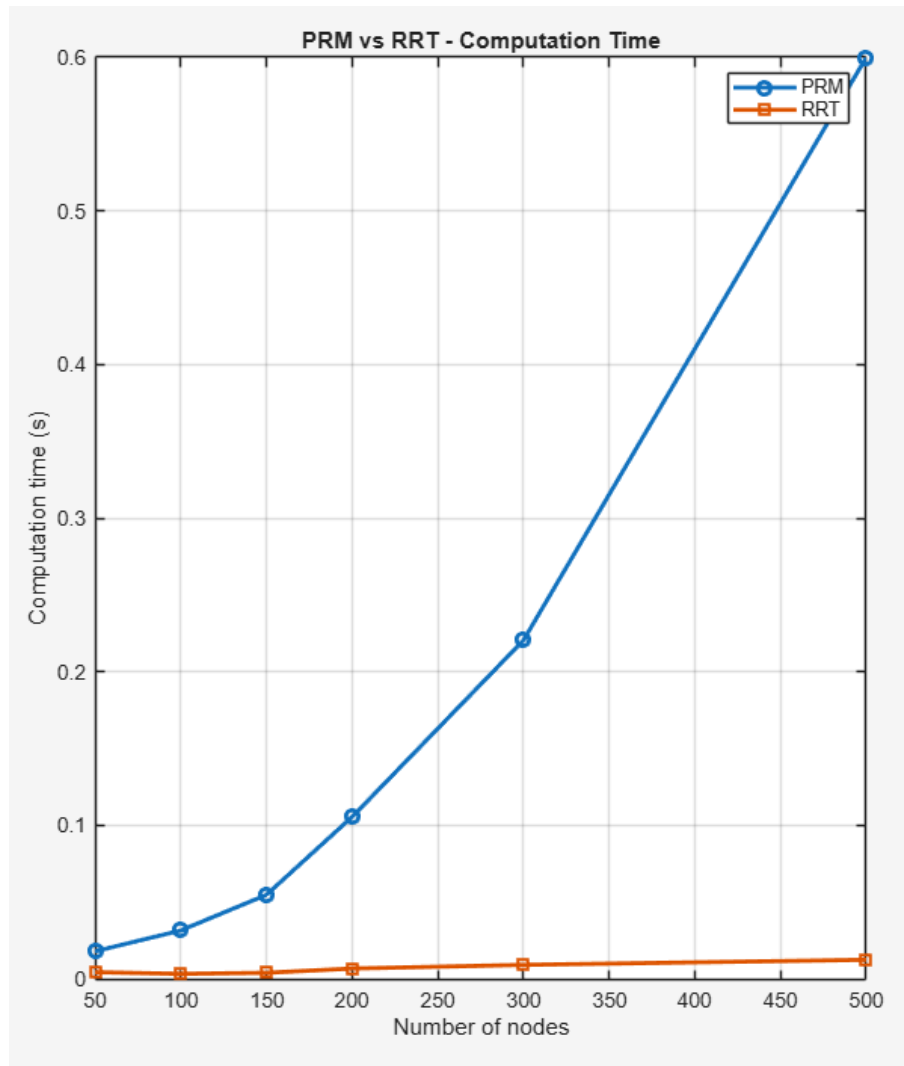


Figure 4: Computation time as a function of the number of nodes

From the graph, PRM becomes significantly slower when the number of nodes increases. This is expected because many possible connections have to be tested. RRT remains much faster in this experiment.

Path Length

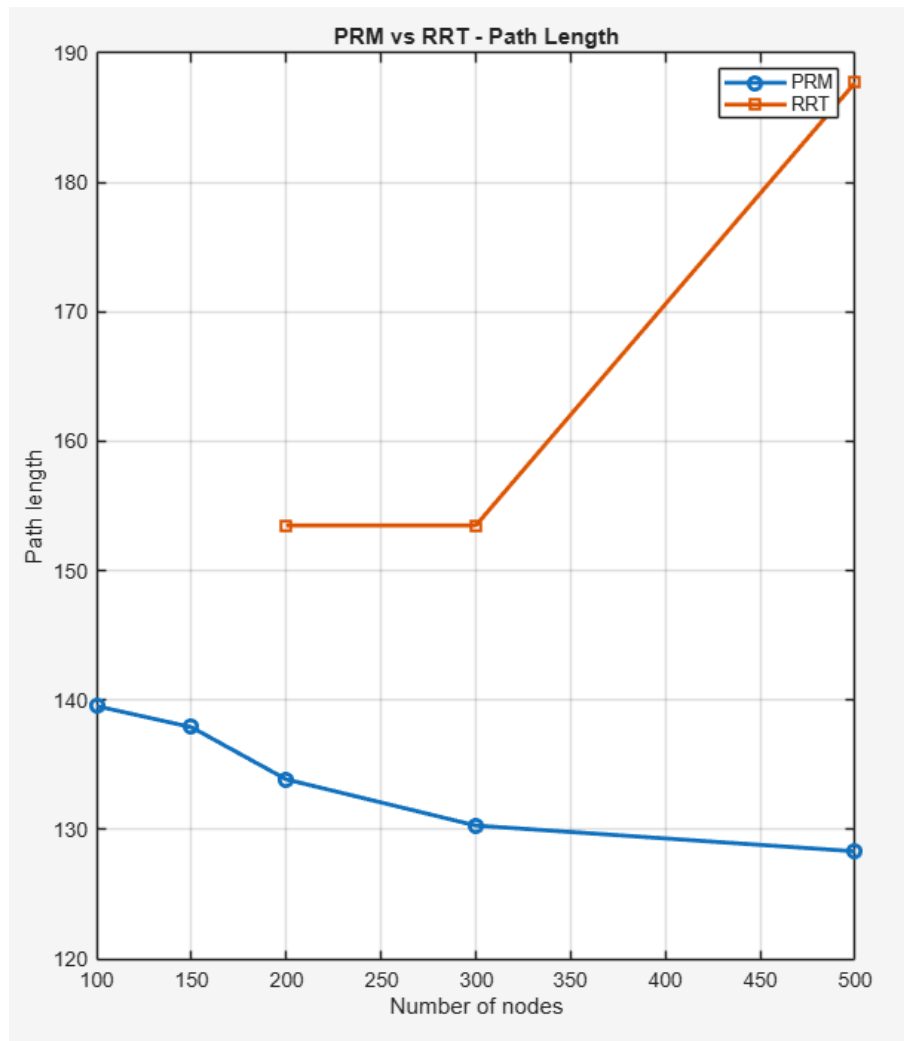


Figure 5: Path length comparison

The PRM path is shorter and more stable. The RRT path is longer because the tree grows in a more random way.

Visual Comparison

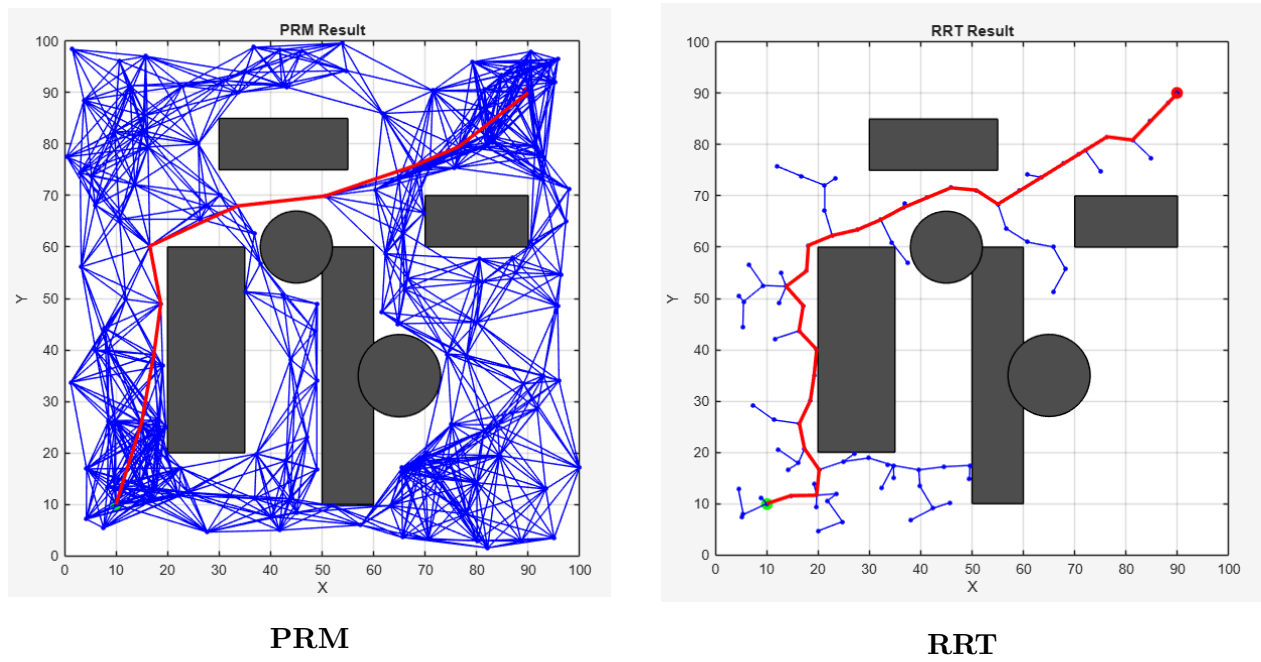


Figure 6: Side-by-side visual comparison of the two algorithms

This final visual comparison confirms the numerical results. PRM gives a cleaner and shorter path, while RRT is faster but less optimized.

Conclusion

In this report, we compared PRM and RRT in the same obstacle map. The results show a clear difference between both methods.

PRM gives better path quality, but its computation time increases quickly with the number of nodes. RRT is simpler and faster, but the obtained path is less efficient.

So, the choice between PRM and RRT depends on the objective: better path quality or faster exploration.